VSB TECHNICAL UNIVERSITY OF OSTRAVA | FACULTY OF MECHANICAL ENGINEERING | DEPARTMENT OF APPLIED MECHANICS

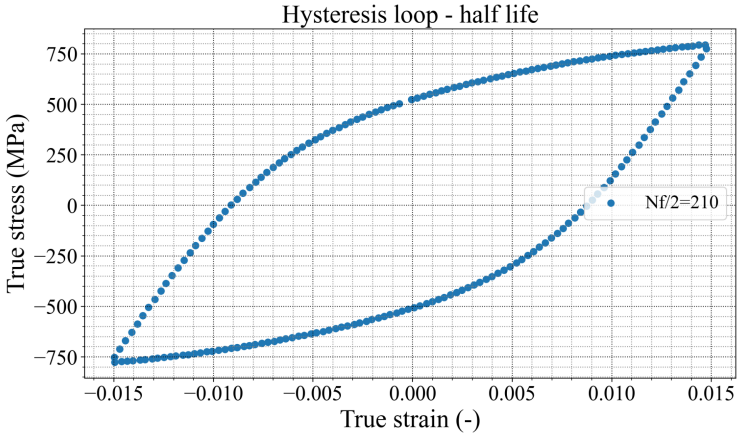
FABER Fatigue Benchmark Repository COST CA23109
cost EUROPEAN COOPERATION IN SCIENCE AND TECHNOLOGY
Funded by the European Union

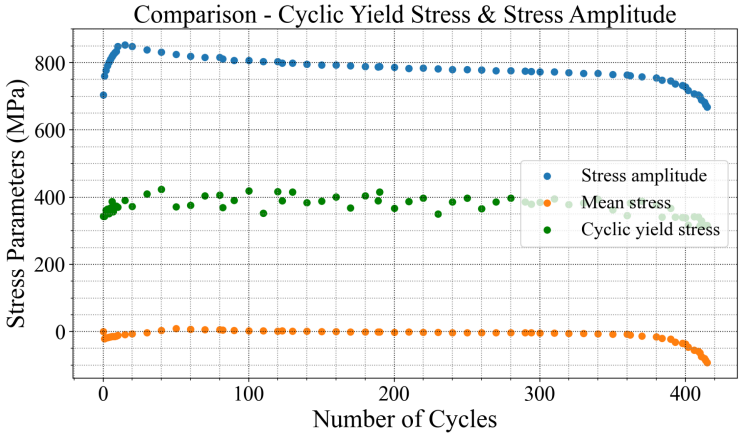
POST-PROCESSING RECORD

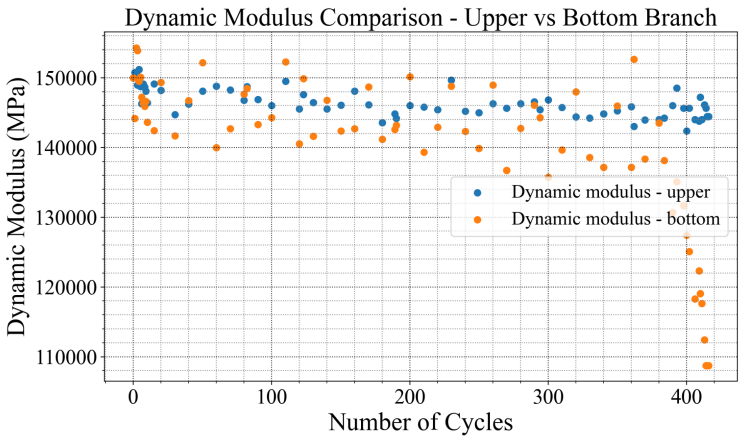
Title:	Uniaxial LCF test report	Date:	9. 6. 2024
Specimen number:	ID4 solid	Dimensions:	6.022 mm
Material:	In718 SLM(Z)	Type of loading:	uniaxial
Equipment:	Instron 8801	Gauge length:	12.5 mm
Institution:	VUT Brno, Czechia	Experimenter:	Josef Zapletal
Temperature:	22°C	Controlled:	Strain-triangular
Amplitude:	1.50 %	Mean value:	0.0%
Loading rate:	0.1 %/s	Unloading rate:	0.1 %/s
Frequency of load.:	0.0333 Hz	Sampling frequency:	200 points/cycle

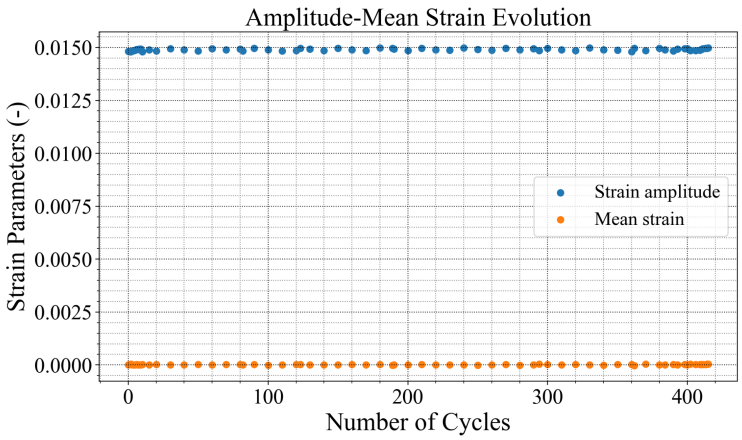
Input data for evaluation

N1: 160	Young modulus E: 151334 MPa	El. region size $\Delta\sigma_{LIN}$: 200 MPa
N2: 330	E - used for Wp: 150000 MPa	Offset pl. strain: 5.000e-04









Half-life values:

ε_a (%)	ε_{ap} (%)	σ_a (MPa)	σ_m (MPa)	N_f (-)	N_f^{E10} (-)	E_{dyn}^{upp} (MPa)	E_{dyn}^{btm} (MPa)	Y_{cyclic} (MPa)
1.492	0.955	786.36	-1.63	416	402	145882.71	143517.65	387.19

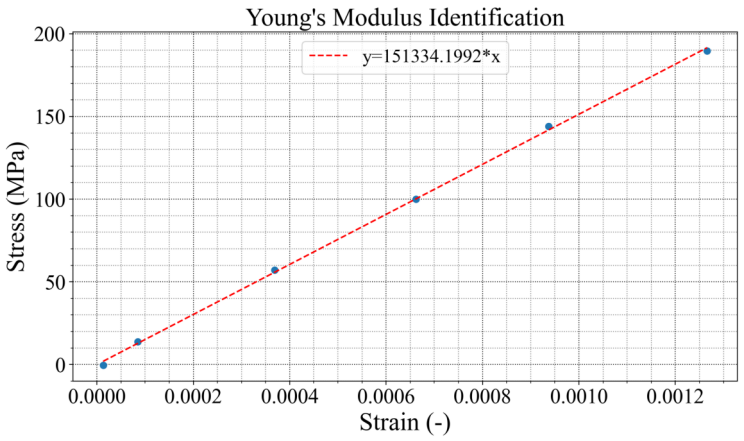
Note: Evaluated with FABER LCF Tool.

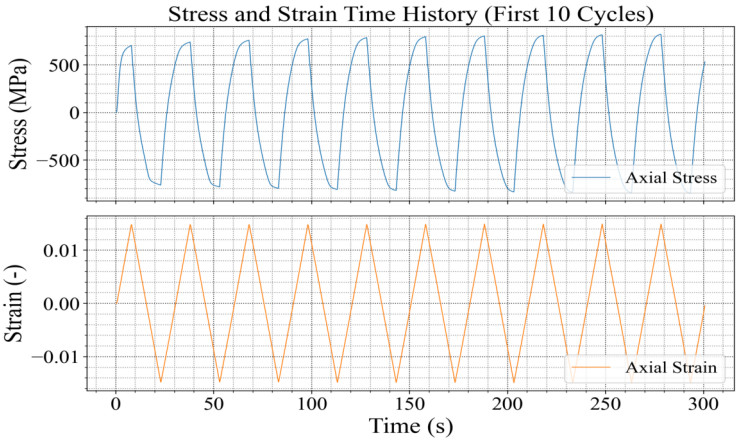
Acknowledgments: Financial support from the Czech Science Foundation under the registration no. 23-05338S.

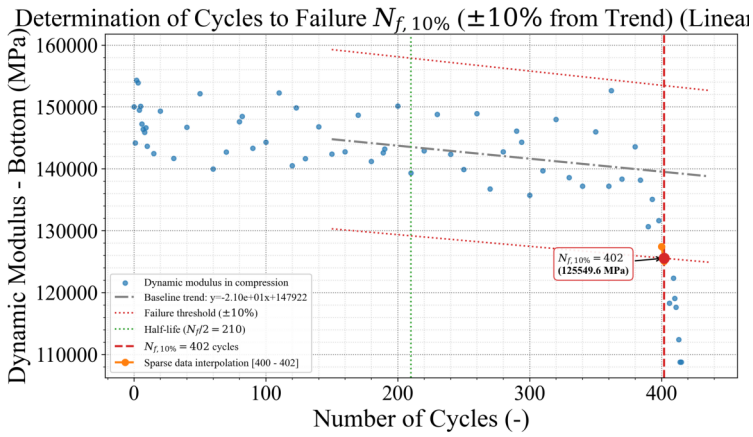
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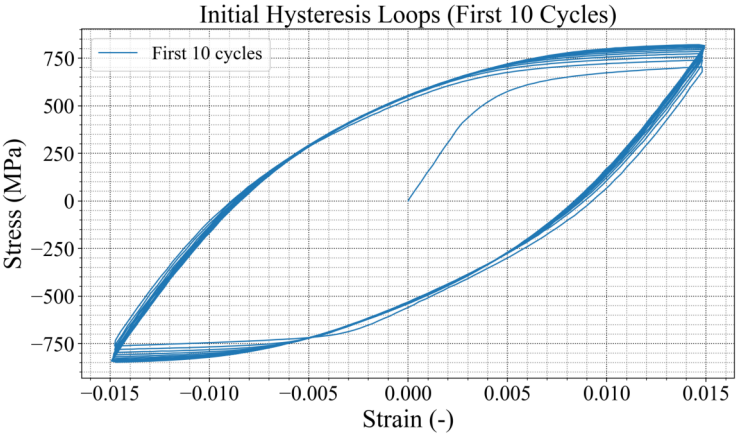
Ref: Tomáš Adamec et al. (2025) "Backstress shift modelling concept for improving uniaxial ratcheting pre...." European Journal of Mechanics, A/Solids 114: 105770

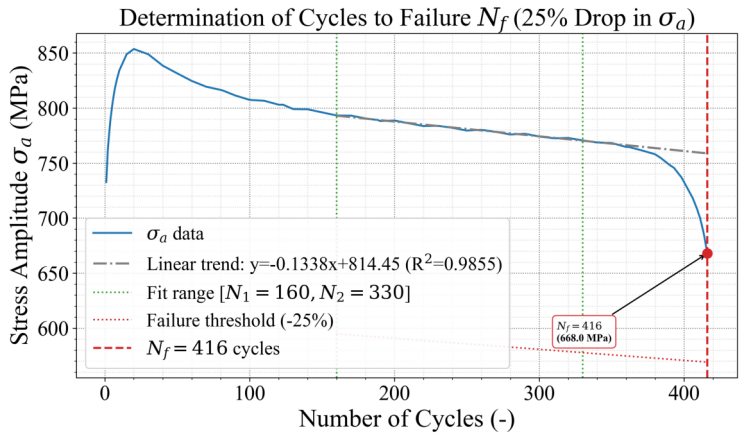
DOI: https://doi.org/10.1016/j.euromechsol.2025.105770

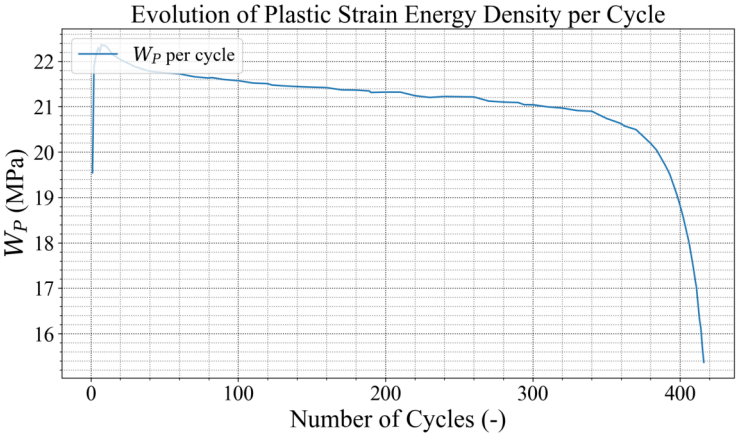












Additional values:

W_p^{tot} (MPa)	ΔW_p^{mean} (MPa)	ΔW_p (MPa)	ΔW_p^{VarE} (MPa)
1384.32	20.661	21.266	21.322